



University
Mohammed VI
Polytechnic



Lab 3: Relational Algebra

Professor: Karima Echihabi
Program: Computer Engineering
Session: Fall 2026 – UM6P – CC

Preparatory Exercises

Before starting Project Part III, students should review and solve the following textbook exercises.

Textbook Exercises

Reference: Ramakrishnan, Raghu, and Johannes Gehrke, *Database Management Systems*, 3rd Ed. [1].

- Chapter 4 (Relational Algebra): Exercises 4.3, 4.5

Objectives

- Express non-trivial queries over RLMS using Relational Algebra (RA).

Tasks

For each query, provide both the Relational Algebra.

1. Find the first and last names of persons who have made at least one equipment reservation.
2. Find the IDs of persons who are either attached to a laboratory or act as PI of at least one research project.
3. Find the IDs of laboratories located in the building 'College of Computing' or containing at least one equipment unit whose model belongs to the category 'CPU Server'.
4. Find the IDs of laboratories that contain at least one equipment unit whose model belongs to the category 'CPU Server' and at least one equipment unit whose model belongs to the category 'GPU Server'.

5. Find the Faculty members who supervise every laboratory located in the building 'College of Computing'.
6. Find the IDs of persons who participate in every research project funded by at least one budget that also funds the laboratory with `labId = 2`.
7. Find pairs of persons (p_1, p_2) such that p_1 has made more reservations than p_2 .
8. Find the IDs of persons who have reserved equipment units located in at least two different laboratories.
9. Find the IDs of reservations submitted in September 2026 that reserve at least one equipment unit currently located in the laboratory named 'AI Movement'.
10. Find the IDs of persons who act as PI of more than one research project.
11. Find the IDs of approvers who have approved reservations involving equipment units located in more than one laboratory.
12. Find the IDs of persons who have no reservation whose planned reservation interval includes November 6, 2026.
13. Find the laboratories whose number of equipment units is below the average number of equipment units per laboratory.
14. For each laboratory, return the person who has made the greatest number of distinct approved reservations involving equipment units located in that laboratory.
15. List the IDs of persons who made at least three equipment reservations during the year 2025.

Deliverable

- A report that: 1) translate the queries into relational algebra expression; 2) summarizes the key challenges faced and how they were addressed.
- The deliverable is due on Sunday 04/10 at 23:59.

References

- [1] Raghu Ramakrishnan, Johannes Gehrke, and Johannes Gehrke. *Database management systems*, volume 3. McGraw-Hill New York, 2003.